1. Pass-by-value:

Statement	Callee a	Callee b	Callee t	Caller (x, y)
start	1	2	-	(1, 2)
t = a	1	2	1	(1, 2)
a = b	2	2	1	(1, 2)
b = t	2	1	1	(1, 2)

x = 1, y = 2

Pass-by-reference:

Statement	Callee a	Callee b	Callee t	Caller (x, y)
start	1	2	-	(1, 2)
t = a	1	2	1	(1, 2)
a = b	2	2	1	(2, 2)
b = t	2	1	1	(2, 1)

x = 2, y = 1

Pass-by-value-result:

Statement	Callee a	Callee b	Callee t	Caller (x, y)
start	1	2	-	(1, 2)
t = a	1	2	1	(1, 2)
a = b	2	2	1	(1, 2)
b = t	2	1	1	(1, 2)
a = x	2	1	1	(2, 2)
b = y	2	1	1	(2, 1)

x = 2, y = 1

2. a.

Pass-by-reference:

Statement	Callee x	Callee y	Caller a
start	10	10	10
x = x + 1	11	11	11
y = y + 1	12	12	12

a = 12

Pass-by-value-result:

Statement	Callee x	Callee y	Caller a
start	10	10	10
x = x + 1	11	10	10
y = y + 1	11	11	10
x	11	11	11

y	11	11	11

a = 11

b. The final values are different because pass-by-reference uses one shared cell. Pass-by-value-result uses separate cells, and it returns the copied values of x and y.

3. When the loop finishes, all 3 closures return 3 because they share the same environment. A static link allows a function to access variables where it was first defined, and since all 3 closures share the same environment, their static links are the same variable. If USILang used a separate cell for i in each loop, that would allow each closure to have its own value. Then, they would return 0, 1, and 2.